

Unraveling the physical properties of pure and lithium thulium iondoped calcium sulfate

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1. INTRODUCTION

Calcium sulfate (CaSO_4) is a highly versatile compound with a wide range of potential applications, mainly due to its ability to exist in several hydration states. These different microstructural forms significantly influence the final properties of the material after dehydration, especially in terms of hardness and mechanical strength. Such features are highly desirable in various applications. Therefore, this work presents a comprehensive DFT investigation of calcium sulfate, aiming to elucidate key physical properties, including lattice parameters, electronic structure, optical response, and mechanical behavior. Through detailed atomic-scale insights, the study seeks to support the development and optimization of CaSO_4 for multifunctional applications ranging from environmental remediation to biomedical engineering and energy technologies.

2. MATERIALS AND METHODS

Calculations were carried out with CASTEP, employing a plane-wave basis set with ultrasoft pseudopotentials and a $6 \times 5 \times 5$ Monkhorst-Pack k-point grid. Structural optimization used the BFGS and LBFGS algorithms, with a cutoff energy of 620 eV and strict convergence criteria for energy, force, pressure, and displacement. Valence electronic configurations were considered for Ca, S, and O atoms. After relaxation, elastic constants were computed with the ELATE program, enabling 3D analysis of mechanical stability and anisotropic elastic properties (Young's modulus, linear compressibility, shear modulus, and Poisson's ratio).

3. RESULTS

The detailed atomic structure in Fig. 1 shows representations of the orthorhombic crystalline structure of CaSO_4 containing sulfate groups of Ca, S, and O bonded to calcium atoms.

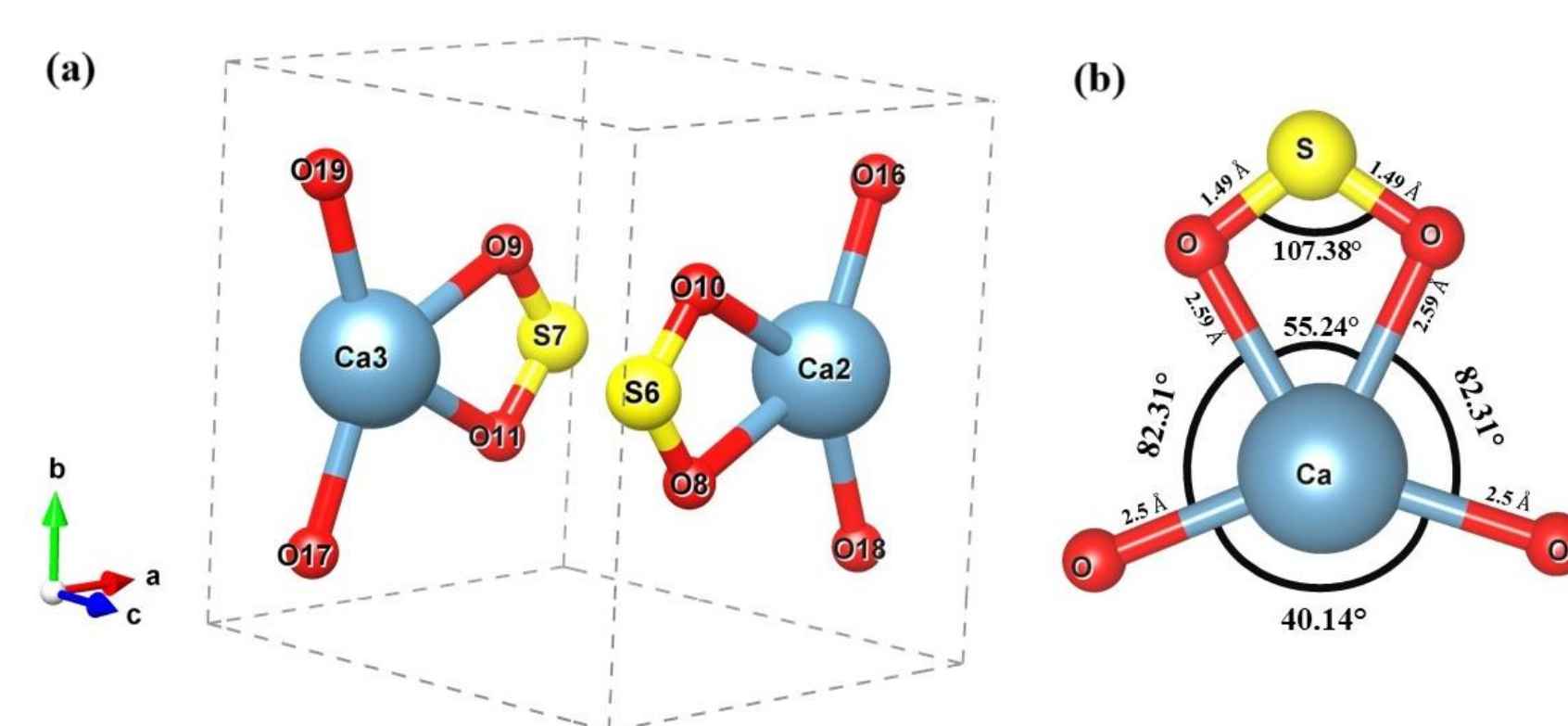


Figure 1: Representation of the crystalline unit cell showing the local coordination geometry of the Ca2 and Ca3 atoms. Source: Santos et al., 2025.

Figures 2(a and b) show the complete electronic band structure and the total density of states (TDOS) of orthorhombic CaSO_4 , estimated via GGA-PBESol.

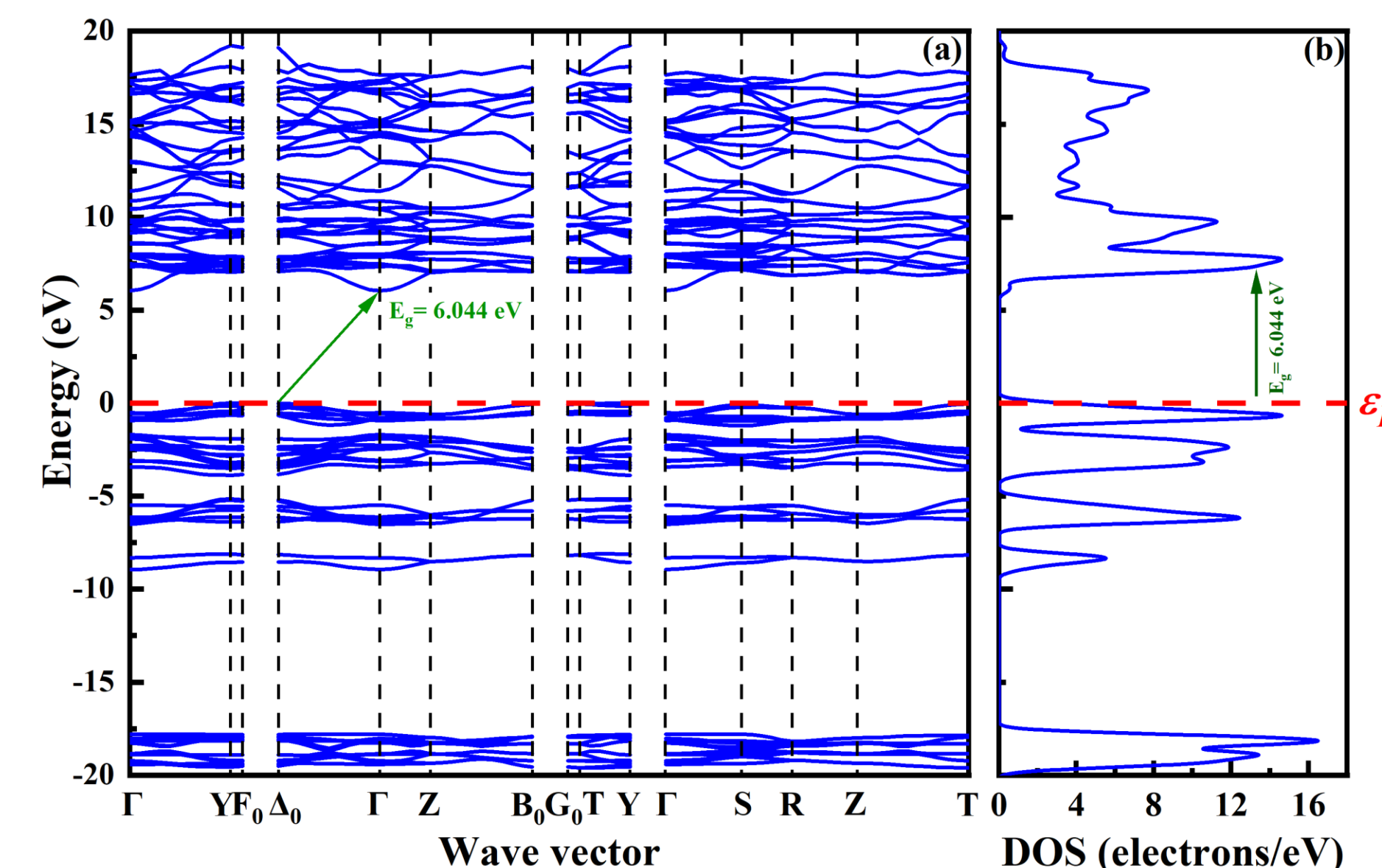


Figure 2: Representation of the crystalline unit cell showing the local coordination geometry of the Ca2 and Ca3 atoms. Source: Santos et al., 2025.

Figure 3 shows the electronic density of states (TDOS) and the projected density of states (PDOS) of orthorhombic CaSO_4 crystal, as a function of energy, expressed in eV, ranging from -20.0 to $+20.0$ eV.

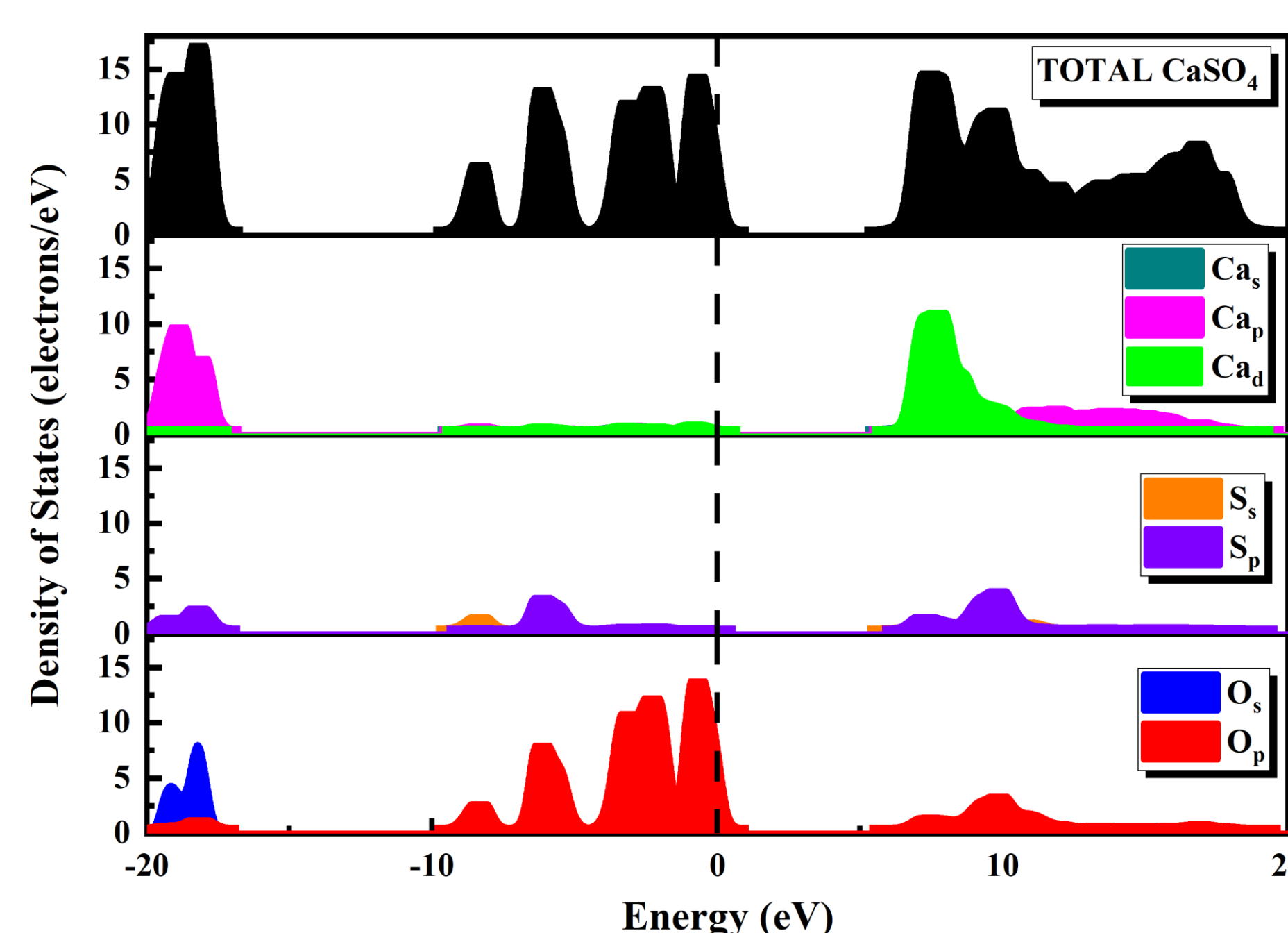


Figure 3: TDOS and PDOS of Ca, S, and O for the energy levels in orthorhombic CaSO_4 . Source: Santos et al., 2025.

Figure 4 shows the optical absorption spectrum of the orthorhombic CaSO_4 crystal, with absorption (in cm^{-1}) represented on the vertical axis and the incident photon energy (in eV) on the horizontal axis.

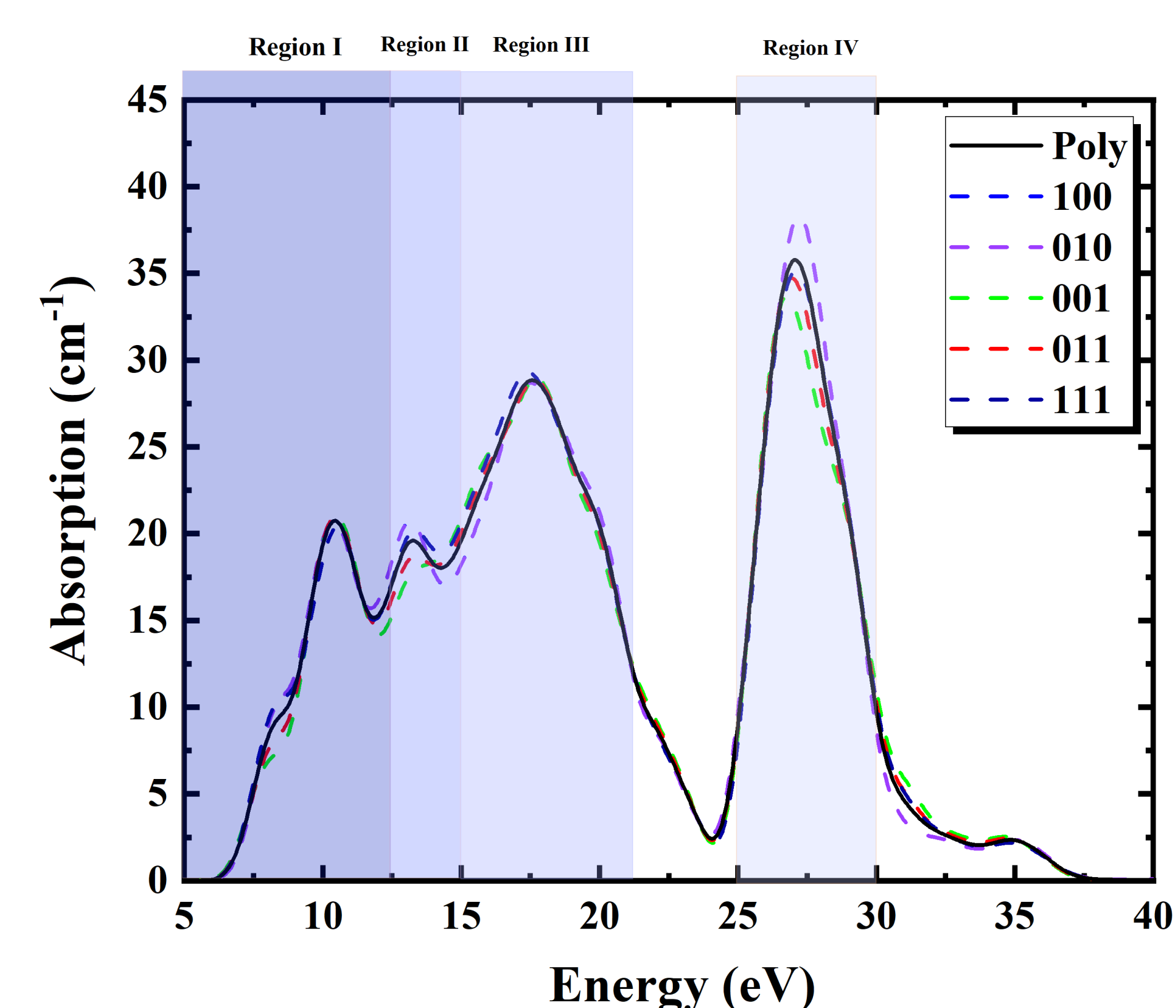


Figure 4: Optical absorptions of orthorhombic CaSO_4 as a function of energy when incident UV light is polarized along the crystalline planes and for a polycrystalline (poly) sample. Source: Santos et al., 2025.

Figure 5(a) represents the complex dielectric function, while Figure 5(b) shows the complex refractive index of orthorhombic CaSO_4 , both plotted as a function of the incident radiation energy (in eV).

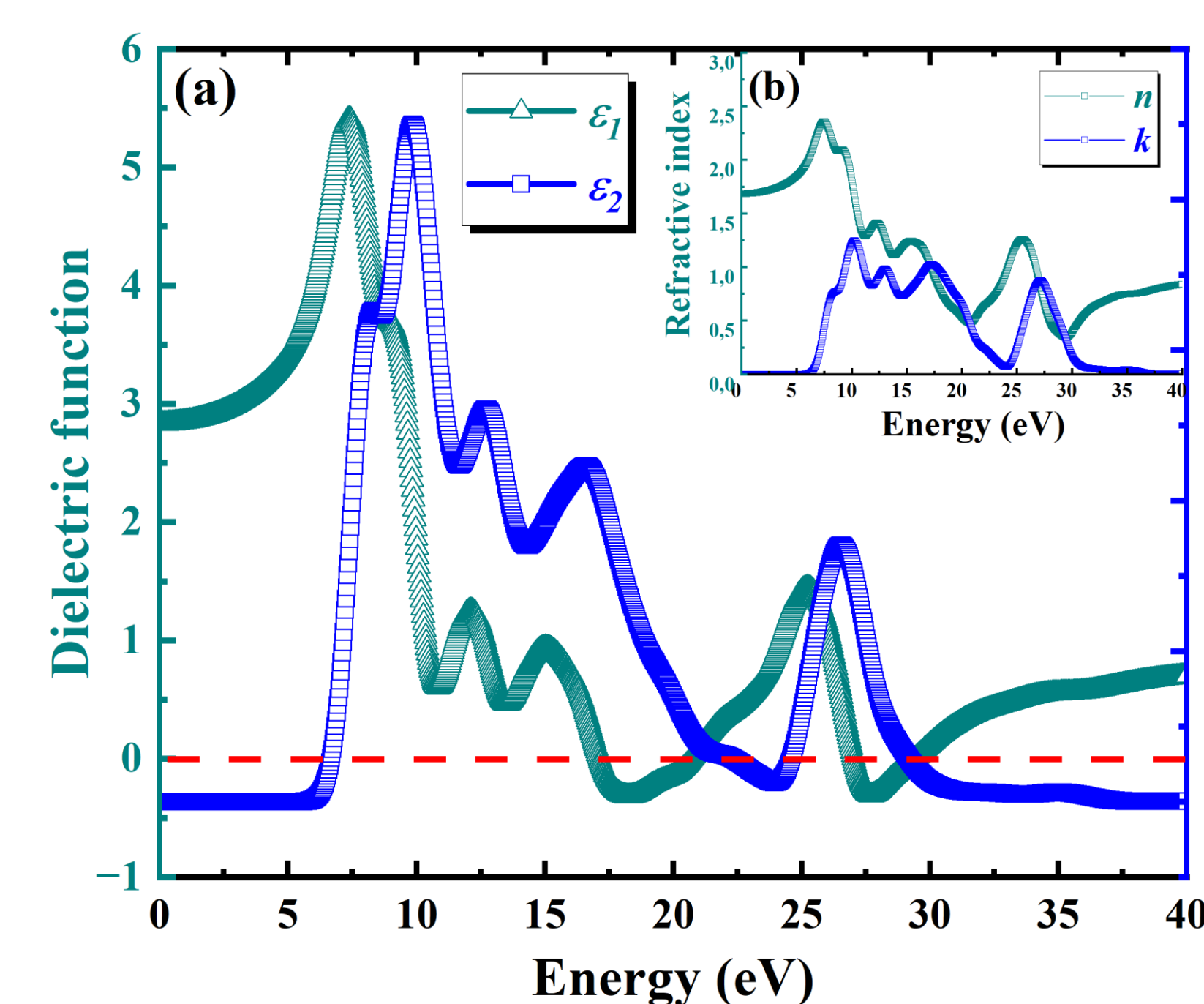


Figure 5: (a) Dielectric function of CaSO_4 . (b) Refractive index of CaSO_4 . Source: Santos et al., 2025.

4. CONCLUSIONS

The study highlights the fundamental properties of CaSO_4 that underpin its potential for diverse technological applications, emphasizing its electronic, optical, and mechanical performance at the atomic scale.